JONATHAN ANDREW SALEH

Data Science Student

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SUMMARY

Data Science student at Bina Nusantara University (GPA 3.71) with three end-to-end machine learning projects shipped as live applications: supervised classification deployed on AWS, semantic search over 57,650 documents, and unsupervised clustering. Comfortable taking a project from raw data to a running inference endpoint.

EDUCATION

Bina Nusantara University 2024 - Present
BSc Data Science Jakarta

GPA: 3.71 / 4.00. Relevant coursework: Machine Learning, Deep Learning, Model Deployment, Data Mining and Visualization, Big Data Infrastructure, Algorithm Design and Analysis.

IPEKA Senior High School 2021 - 2024
High School Diploma Balikpapan

Valedictorian, ranked 1st of the entire parallel batch.

PROJECTS

Credit Score Classification: End-to-End ML Deployment Machine Learning
Python, scikit-learn, MLflow, AWS SageMaker, EC2, Streamlit Code | Live Demo

- Built an object-oriented machine learning pipeline classifying 25,000 customer records into three credit tiers across 28 financial features, reaching a weighted F1 of 0.73.
- Compared Random Forest, Hist Gradient Boosting, and Extra Trees in MLflow, then tuned the winner across 12 hyperparameter configurations with 3-fold cross-validation, scoring on weighted F1 rather than accuracy to handle class imbalance.
- Deployed the model to AWS SageMaker as a real-time inference endpoint, called from a Streamlit app on EC2 through boto3.

Song Search: Multi-Method Search Engine NLP / Semantic Search
Python, sentence-transformers, TF-IDF, fuzzy matching, Streamlit Code | Live Demo

- Built a search engine over 57,650 songs by 643 artists, running four retrieval methods from a single query bar: title substring, fuzzy artist matching, lyric phrase match with TF-IDF ranking, and semantic search.
- Precomputed 384-dimensional sentence embeddings (all-MiniLM-L6-v2) for every song, so semantic queries resolve without encoding at request time.
- Tuned fuzzy matching with a grid search on a labelled test set, selecting the scorer and a similarity cutoff of 75.

NBA Roster Lab: Player Archetypes with Unsupervised ML Unsupervised Learning / Clustering
Python, scikit-learn, pandas, Streamlit, Vercel Code | Web App | Streamlit

- Clustered 9,943 player-seasons (2000-2026) into seven playing-style archetypes with k-means on eight rate-based features, deliberately excluding quality metrics so the clusters capture style rather than skill.
- Selected k = 7 on interpretability after the elbow and silhouette methods gave no sharp answer, validating the clusters against recognisable players.
- Built four tools on the archetypes: statistical twins via cosine similarity, a roster diagnostic benchmarking teams against era-winning rosters, a custom roster builder, and league-evolution charts.

EXPERIENCE

Assistant Aug 2024 - Present
Kumon Balikpapan

- Review and grade worksheets for ~10 students daily across Kumon's Mathematics and English tracks, covering students enrolled in one or both subjects.
- Grade across multiple curriculum levels, where worksheet volume and difficulty differ per student, and report recurring error patterns to supervisors so weak spots are corrected before a student advances a level.

Sales & Advisory Associate Dec 2024 - Feb 2025

Allianz Indonesia Jakarta

- Completed Allianz's agency training on needs-based financial analysis and product structuring.

ORGANIZATION EXPERIENCE

Event Division 2025 - Present

Data Science Club, Bina Nusantara University Jakarta

- Organize club events as event organizer, including a bonding event for incoming and existing activists and an ongoing badminton tournament.
- Handle logistics for Outventure, the club's outbound event, and for an upcoming seminar, covering food, equipment, and property procurement.

Drummer - Music Team 2022 - 2025

GKY Greenville Church Jakarta

- Played weekly services as a drummer on the church music team for three years.

SKILLS

Languages	Python, SQL
ML & Data	scikit-learn, pandas, NumPy, classification, k-means clustering, sentence embeddings, TF-IDF
Tools	Git, MLflow, Streamlit
Cloud	AWS (SageMaker, EC2)

LANGUAGES

Bahasa Indonesia - Native / Professional Proficiency

English - Full Professional Proficiency

ACHIEVEMENTS

- **Top 20 (Ranked 15th), AI Innovation Challenge (2026):** COMPFEST 18, preliminary round.
- **Valedictorian, 1st Rank (2024):** Graduated first in the entire parallel batch from IPEKA Senior High School.
- **Gold Medalist, English Category (2023):** National High School Olympiad (NHSO).
- **3rd Place, English Speech Competition (2023):** Organized by the Balikpapan English Teachers' Association (MGMP).

INTERESTS

Tennis - Recreational player

Music - Drums